

Alexander Bernstein

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Education

University of California, Santa Barbara

Ph.D. in Statistics

Santa Barbara, CA

Sept. 2025

Dissertation: “Long-Only Minimum Variance Portfolio Composition for Factor Models”

Washington University in St. Louis

M.Sc. in Systems Science and Applied Mathematics

St. Louis, MO

Dec. 2016

B.A. in Mathematics and Economics (Cum Laude)

May 2014

Technical Skills

Quantitative Methods: Convex & Portfolio Optimization, Factor Models, Asset Pricing (options & bonds), Black-Litterman, Risk Modeling, Time Series Analysis, Stochastic Control, Reinforcement Learning, Asymptotic Theory.

Programming & Tools: Python (NumPy, SciPy, Pandas, Scikit-Learn, PyTorch, CVXPY), R, MATLAB, SQL, C, L^AT_EX, Git, Linux, AWS.

Selected Projects

Factor-Model Portfolio Pipeline: Spec-driven **Python** pipeline for S&P 500 portfolio construction. **PCA** factor models with pluggable covariance estimators (**POET** thresholding, eigenvector shrinkage, GARCH) and return estimators (**Kalman**/EKF filters, robust Student-t, Black-Litterman, L1-adaptive control); constrained mean-variance optimization via **CVXPY** (CLARABEL); weekly walk-forward backtest over a 260-week rolling window (10 PCA factors) on the S&P 500, with a live rebalancer.

Quantitative Options Pricing:  Implemented **Monte Carlo** (Euler/Milstein) and **finite-difference** (incl. Crank-Nicolson) solvers for the **Heston** and **CEV** models; built **Gaussian Process** (Matérn) and **PyTorch** neural-network surrogates for fast pricing and **Greeks** via autograd.

Conditional Covariance Estimation:  Built a **Covariance Regression** model for covariate-adjusted covariance of idiosyncratic equity returns — a parsimonious alternative to **MGARCH** that improves stability and cuts parameter count in high dimensions.

Mean Field Game Reinforcement Learning:  Implemented a **Q-Learning** solver for the major-minor cyber-security mean field game (Carmona & Delarue, 2018), learning optimal defense policies against an evolving population distribution μ .

Research

Publications:

Banerjee, T., **Bernstein, A.**, Feinstein, Z. (2025). “Dynamic clearing and contagion in financial networks.” *European Journal of Operational Research* 321(2), 664–675.

Bernstein, A., Shkolnik, A. (2025). “Asymptotics of Quadratic Forms on a Simplex.” *In preparation*.

Selected Talks: INFORMS Annual Meeting (2023); SIAM Conf. on Applied & Computational Discrete Algorithms (2021, poster); CDAR Risk Seminar, UC Berkeley (2020).

Experience

University of California, Santa Barbara

Teaching Assistant & Graduate Student Mentor

Santa Barbara, CA

Sept. 2017 – June 2025

Taught statistical theory, stochastic processes, regression, time series, financial mathematics, and risk theory; mentored undergraduate research in financial mathematics and optimization.

Epic Systems

Technical Services Engineer

Madison, WI

Sept. 2014 – Sept. 2015

Tuned the Epic EHR codebase for large institutional clients; developed fixes for high-availability and regulatory compliance.

Prozess Technologie

Intern

St. Louis, MO

May 2013 – May 2014

Built computational simulations to validate laboratory equipment and calibrated pharmaceutical manufacturing hardware to industry standards.